Control & Coordination — Short Answers

Class X Biology · Concise, grammatically complete answers · Print-friendly

PART 1 · Neuron, Synapse & Reflex Arc

Q1. Structure of a neuron — parts (a) & (b). 2 m

(a) Dendrites are the branched parts that receive information. (b) The axon is the long fibre along which the electrical impulse travels. Draw and label both in your diagram.

Q2. What is a synapse? Why is there a delay? 2 m

A synapse is the small gap between two neurons. There is a brief delay because the neurotransmitter has to be released, diffuse across the gap, and bind to the next neuron.

Q3. What happens at the synapse? 2 m

At the synapse, the electrical impulse releases neurotransmitters into the gap. They diffuse across and bind to receptors on the next neuron, generating a new impulse.

Q4. Fill (A) & (B) in the reflex arc. 2 m

(A) is the sensory neuron, which carries the impulse from the receptor to the spinal cord. (B) is the motor neuron, which carries it from the spinal cord to the effector.

Q5. Why is the signal processed in the spinal cord? 2 m

Reflexes are processed in the spinal cord so that the response can be immediate. Sending the signal to the brain and back would take longer and delay the protective action.

Q6. Neuron unit, electrical & chemical signal, backward flow. 3 m

(a) The neuron is the structural and functional unit of the nervous system. (b) An electrical impulse travels through the axon, and at its end it is converted into a chemical signal, as neurotransmitters carry the message across the synapse. (c) The impulse cannot flow backward because neurotransmitters are released only at the axon terminal and receptors are present only on the next neuron.

Q7. Reflex action vs involuntary action. 3 m

A reflex action is a quick, automatic response to a stimulus, controlled by the spinal cord, for example withdrawing the hand from a hot object. An involuntary action is a routine, brain-controlled activity such as the heartbeat or breathing.

Q8. Touching a hot object. 3 m

(a) Thermoreceptors in the skin detect the heat. (b) The spinal cord is the main integration centre. (c) The skeletal arm muscles contract and pull the hand away.

PART 2 · Human Brain & CNS Protection

Q9. How is the CNS protected? 2 m

The brain and spinal cord are protected by the bony skull and vertebral column, by the three meninges, and by the cerebrospinal fluid, which acts as a shock absorber.

Q10. Balance lost after alcohol — which part is affected? 2 m

The cerebellum is affected. It maintains posture and body balance, and it also coordinates precise voluntary muscle movements.

Q11. Match brain parts to activities. 3 m

(a) Cerebellum — balance. (b) Medulla — blood pressure, salivation, vomiting. (c) Hypothalamus — hunger and feeling full. (d) Cerebrum — thinking and intelligence. (e) Medulla — respiration. (f) Midbrain — reflex movements of the head, neck and trunk.

Q12. Brain diagram + why medulla injury is fatal. 5 m

(a) Draw a labelled diagram showing the forebrain, midbrain, hindbrain, cerebrum, cerebellum, medulla and pituitary gland. (b) The medulla controls the vital functions of heartbeat, breathing and blood pressure, so injury to it stops these processes and death occurs immediately.

PART 3 · Plant Hormones & Tropic/Nastic Movements

Q13. Phototropism and geotropism. 2 m

Phototropism is the growth of a plant part in response to light, and geotropism is its growth in response to gravity. Shoots grow upward (negative geotropism) while roots grow downward (positive geotropism).

Q14. Why does Mimosa fold on touch? 2 m

When touched, cells in the pulvinus at the base of the leaves lose water and become flaccid, so the leaflets fold. This is a nastic movement caused by change in turgor pressure, not by nerves.

Q15. How do pea tendrils climb? 2 m

On touching a support, the tendril cells on the contact side grow slower while those on the opposite side grow faster, so the tendril coils around it. This is thigmotropism.

Q16. Match plant hormones. 3 m

Auxin causes cell elongation and phototropic curvature; gibberellins cause stem elongation and seed germination; cytokinins promote cell division; abscisic acid inhibits growth and closes stomata.

Q17. Tropic vs nastic movements. 3 m

Tropic movements are slow, irreversible, growth-based responses directed toward or away from the stimulus, like phototropism. Nastic movements are fast, reversible turgor changes whose direction is independent of the stimulus, like the folding of Mimosa.

PART 4 · Endocrine System & Feedback

Q18. Why are they called ductless glands? 2 m

Endocrine glands are called ductless because they have no ducts. Their secretions, called hormones, are released directly into the blood, which carries them to the target organs.

Q19. Emergency hormone. 2 m

The emergency hormone is adrenaline, secreted by the adrenal glands. During fight-or-flight it increases the heart rate and raises the blood-sugar level to supply extra energy.

Q20. Hormone & gland for each condition. 3 m

(a) Goitre — thyroxine from the thyroid gland. (b) Gigantism — growth hormone from the pituitary. (c) Diabetes — insulin from the pancreas. (d) Underdeveloped secondary characters — testosterone from the testes.

Q21. Feedback mechanism — blood sugar. 3 m

When blood sugar rises, the pancreas secretes insulin, which lowers it; when it falls, it secretes glucagon, which raises it. This negative feedback keeps the blood-sugar level balanced.

Q22. Nervous vs endocrine + why hormonal control. 5 m

(a) The nervous system is fast, precise and short-lived, through impulses along neurons; the endocrine system is slower and long-lasting, carrying hormones through blood. (b) Chemical control is necessary because nerve impulses do not reach every cell, neurons need time to reset, and hormonal signals reach all cells and sustain long-term growth and metabolism.